

1-Stick

BENTELER Steel/Tube
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qms.shreveport@benteler.com

BENTELER
Steel/Tube

INSPECTION CERTIFICATE EN 10204-3.1

EN 10204:2005-01

Kelly Pipe Co., LLC
PO Box 2827
SANTA FE SPRINGS CA 90670
USA

Document No.: 465-1490280/001/P
Purchase Order No.: 10P44479
Benteler Order No.: 48010476
Dispatch Note No.:
Product: SEAMLESS STEEL TUBES

Inspection No.: Page: 1 / 3
Manufacturer: BSTMC Shreveport USA
Manufacturer's brand: 
Stamp of the inspection representative: MI
Steelmaking process: ELECTRIC FURNACE

Terms of delivery: ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175/ISO 15156-1: 2020, ANSI/NACE MR0175/ISO 15156-2: 2020, ANSI/NACE MR0103/ISO 17945:2015

Dimensions-tolerances: ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, ASTM-A 53-2020, ASME SA-53, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018

Steel grade: GRADE B, GRADE C, GRADE BN, GRADE X42N

Delivery condition: N

Product marking: S: BENTELER - BENTELER-Sign - API 5L 1072 - API-Sign - Month/Year - A/SA 53 A/SA 106 NACE - 4,500" OD x 0,337" WT - B/C/BN/X42N - PSL2 - SMLS - Heat No. - Tube length feet - 2.970 - 15037010 - 10P44479 - LONGHORN

AEZ = Etching ink marking FK = color marking FS = paint stenciling FSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PKE = tag marking PS = die stamp TS = Ink jet spray marking

Item	Dimensions	Heat No.	Test pressure	
			PSI	sec
0007	4 NPS - SCH 80 35,000 FT - 45,000 FT	319126	2970	5

Country of melt and pour

Melted & Manufactured in USA
Nucor, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0007	319126	0.180	0.262	1.00	0.012	0.005	0.17	0.03	0.08	0.18	0.003	0.001	0.002	0.0000

- Formula: $CE\ IIW = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0.43\ \%$
- Formula: $Cr + Cu + Mo + Ni + V \leq 1.00\ \%$
- Formula: $Nb + V \leq 0.06\ \%$
- Formula: $Nb + Ti + V \leq 0.15\ \%$

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Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0007	319126	0,405	0,463	0,004	0,006

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0007	319126	0.170	0.250	0.95	0.012	0.004	0.17	0.03	0.08	0.18	0.004	0.002	0.003	0.0002
0007	319126	0.180	0.250	0.94	0.011	0.004	0.16	0.03	0.08	0.19	0.004	0.002	0.003	0.0002

- Formula: $CE_{IIW} = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0.43 \%$
- Formula: $Cr + Cu + Mo + Ni + V \leq 1.00 \%$
- Formula: $Nb + V \leq 0.06 \%$
- Formula: $Nb + Ti + V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0007	319126	0,386	0,464	0,006	0,009
0007	319126	0,393	0,464	0,006	0,009

Test results

The tubes have been submitted to a leak tightness test by: Hydrostatic test: acc. to API 5L; Test pressure/Time-record, the test records/charts shall be available for examination by the inspector PASSED

The tubes are non destructive tested: ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0,125 IN.; ET - Notch shape/hole: drilled hole PASSED

Visual inspection: PASSED Dimensions examination: PASSED Flattening test: PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0,5/Rm
Requirements			IN	RT 0,5 % PSI 42100-71800	Rm PSI 70000-95000	A2" %		MAX 0,93
0007	000011	319126	1,000 X 0,329	52794	76435	31,00	22,00	0,69

Elongation formula

- Formula $AF = 625000 * AXC^{0,2} / U^{0,9}$

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Notched bar impact test [1 CHARPY_V]

Specimen position: longitudinal (L); Test temperature: -50 °F

Item	Specimen No.	Heat No.	Specimen dimensions			Absorbed energy		Impact strength	Shear fracture	
Requirements			Length	Width	Height	single	average		single	average
			mm	mm	mm	ft-lbf	ft-lbf		%	%
			55	6,67	10,00		MIN 000			
0007	000011	319126	55	6,67	10,00	119			90	
				6,67	10,00	129			90	
				6,67	10,00	117			85	88

Protection

outside surface varnished

Item	Heat No.	Production No.
0007	319126	15037010

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156: HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945: HRC max. 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Data converted from SI units, No weld repair has been carried out

Plant 465, 1 BENTELER DRIVE, LA 71115 Shreveport, 28.10.2024, TEL.: 0318-216-4139

Inspection representative

SH

We certify that the supplied product has been processed, sampled, tested, and inspected in accordance with the agreed specifications and/or references mentioned herein and meet the requirements for the grade indicated. This document was prepared by means of electronic data processing and is valid without signature.

restricted